

Why can't AI companies use salt water instead of fresh water?

Posted by u/JoesBowie • 0 points • 19 comments

Why aren't they even trying to make AI ethical? Why is there no laws preventing the expansion of AI?

Comments

u/StraightAirline8319 • 2 points • 2025-10-09 04:53 UTC

You ever see snowy places and they use salt and other mixtures on their roads? You can see rusty metal though around where they use it?

u/Ticklememoistaint • 2 points • 2025-10-09 04:26 UTC

Because they don't want their AIs to be salty.

u/Delicious_Toad • 2 points • 2025-10-09 04:23 UTC

In addition to being corrosive, it creates scaling issues.

Data centers frequently use evaporative cooling. Salt water is unsuitable for evaporative cooling because it's full of solutes. As it evaporates, the solutes can condense out of solution, forming a rind of crystallized solutes that have to be cleaned out. That's 'scaling.'

Even fresh water leaves behind some gunk, so trays and other components do need to be cleaned regularly, but if you used saltwater you'd have to clean the equipment way more frequently. And, as others have mentioned, even with frequent cleaning they would have a shorter service life because they'd corrode much faster.

u/[deleted] • 1 points • 2025-10-09 04:22 UTC

[removed]

u/Darian_Kimberly • 2 points • 2025-10-09 04:22 UTC

is the water just dumped downstream after or is it piped back to the treatment plant? genuine question

u/Delehal • 2 points • 2025-10-09 06:34 UTC

Depends on the setup. A lot of data centers use evaporative cooling, meaning the water evaporates into the atmosphere. If you used seawater for evaporative cooling, it would leave behind large amounts of salt or brine, which can become a toxic waste problem if it's done at industrial scales.

There are some data centers that are designed to use seawater and release it back into the sea, though. It's just not a common setup for several reasons such as cost and location.

u/ttlanhil • 2 points • 2025-10-09 05:12 UTC

Both - water is used to take a lot of heat energy out of the centre, so depending on your supply or water (and climate) you might cycle it, or just use some as it goes past

u/Bobbob34 • 0 points • 2025-10-09 04:09 UTC

Why is this asked multiple times a day? Salt water is insanely destructive.

u/JoesBowie • 0 points • 2025-10-09 05:26 UTC

Sorry I never saw them

u/programmerOfYeet • 2 points • 2025-10-09 04:04 UTC

This sub needs to start banning duplicate posts

u/BrokenHandBones • 0 points • 2025-10-09 04:11 UTC

"My /new feed needs to be perfection for me to be happy"

u/ttlanhil • 2 points • 2025-10-09 04:03 UTC

Dude, pick one question - the title and text are totally different things

u/BrokenHandBones • 2 points • 2025-10-09 04:10 UTC

People on Reddit will complain about anything. If OP made two posts he'd catch shit too, can't win.

u/RusticBridge • 2 points • 2025-10-09 04:06 UTC

They're related.. fresh water is a limited resource that humans need to survive. AI while a potential boon for society one day is merely a luxury that doesn't really provide benefit to the majority of people at this point

u/ttlanhil • 2 points • 2025-10-09 05:10 UTC

A question about why data centres need fresh water is a reasonable one.
And not too hard to answer from a factual/economic standpoint.

Why companies need huge data centres, similar. There are solid answers.

Either of those would be fine.

Why something like an "AI company" even exists (or people thinking "AI" exists, the LLMs of today came out of that research area, but they aren't intelligent), why they need huge processing centres - also answerable, but needs more detail.

Why companies don't act ethically?

Very different question, and you can approach it from an economic standpoint, but also needs a policy/legal viewpoint.

Why no laws limiting it? Well, there are, but even if there weren't - that's a govt policy/legal sort of question that's going to be ****quite**** involved

Either of those, potentially fine but you need to be pretty specific about the question to get an answer that you want

u/flingebunt • 2 points • 2025-10-09 04:01 UTC

Saltwater is very corrosive so better to stick with freshwater for cooling.

u/Pastadseven • 5 points • 2025-10-09 04:00 UTC

Salt water's corrosive as fuck. Same reason we dont use it in...much of anything.

u/RusticBridge • 2 points • 2025-10-09 03:59 UTC

Al is where the money is and most governments can be bought pretty cheaply

u/MashTactics • 9 points • 2025-10-09 03:59 UTC

Salt is highly corrosive.

u/BrokenHandBones • 2 points • 2025-10-09 04:09 UTC

I assume any material that would resist salt corrosion would be too expensive to use to justify it too.